

Jingzhang Human City OS: From an AI Showcase to a City for People

v1.0: Turn “unconfirmed accountability” into an auditable startup protocol

This iteration adds `governance-startup-protocol.json`, which splits pre-authorization research or pilot startup into five rejectable and reversible stages: scope freeze and startup registration, eligibility and conflict review, public notice and objections, human accountability and authorization record, and version handoff/review/exit. The protocol covers 6/6 readiness scenarios, 3/3 release-gate records, and all 8/8 required fields while explicitly keeping `authorized=false` and `operatorconfirmed=false`; no operator, institution, permit, or field result is named. [data:visual/assets/governance-startup-protocol.json] [data:visual/assets/pilot-readiness-register.json] [depth:phasingimplementation]

The network-free `check-governance-startup-protocol.js` resolves 5/5 stages, 6/6 scenarios, 3/3 release gates, 8/8 fields, four role-selection rules, and five no-go conditions. Two negative fixtures—missing selection inputs and an unauthorized stage—must fail. It proves only that startup conditions and claim boundaries are reviewable inside the package; it does not prove authorization, staffing, field operation, or performance. [data:visual/assets/check-governance-startup-protocol.js] [metric:versionreleasecount]

The new `ai-design-exploration-log.json` records five key choices as “AI-assisted structuring/replay → human judgement → retained or rejected”: release gates, the ordinary-person service chain, spatial parameter trade-offs, the handoff receipt, and the startup protocol. Each returns to existing evidence and still requires a human decision. This is a retrospective structural index, not a verbatim model transcript, external-network log, or field experiment; missing evidence remains conceptual or `unknown`. [data:visual/assets/ai-design-exploration-log.json] [data:visual/assets/check-ai-design-exploration-log.js] [depth:riskmissingdata]

Together these assets put “who may start, who may stop, and whose decision retained the trade-off” on one auditable chain without turning a governance gap into a completion claim. All spatial content remains a conceptual suggestion/reference scheme for professional teams to deepen after official boundaries, rights, controls, existing conditions, mobility, utilities, energy, hydrology, and public baselines are available; formal metrics and provisional geometry remain unchanged. [data:geometry/siteboundary.geojson#SITE-001] [metric:siteareasqm]

v0.9: Turn conceptual actions into a replayable handoff receipt

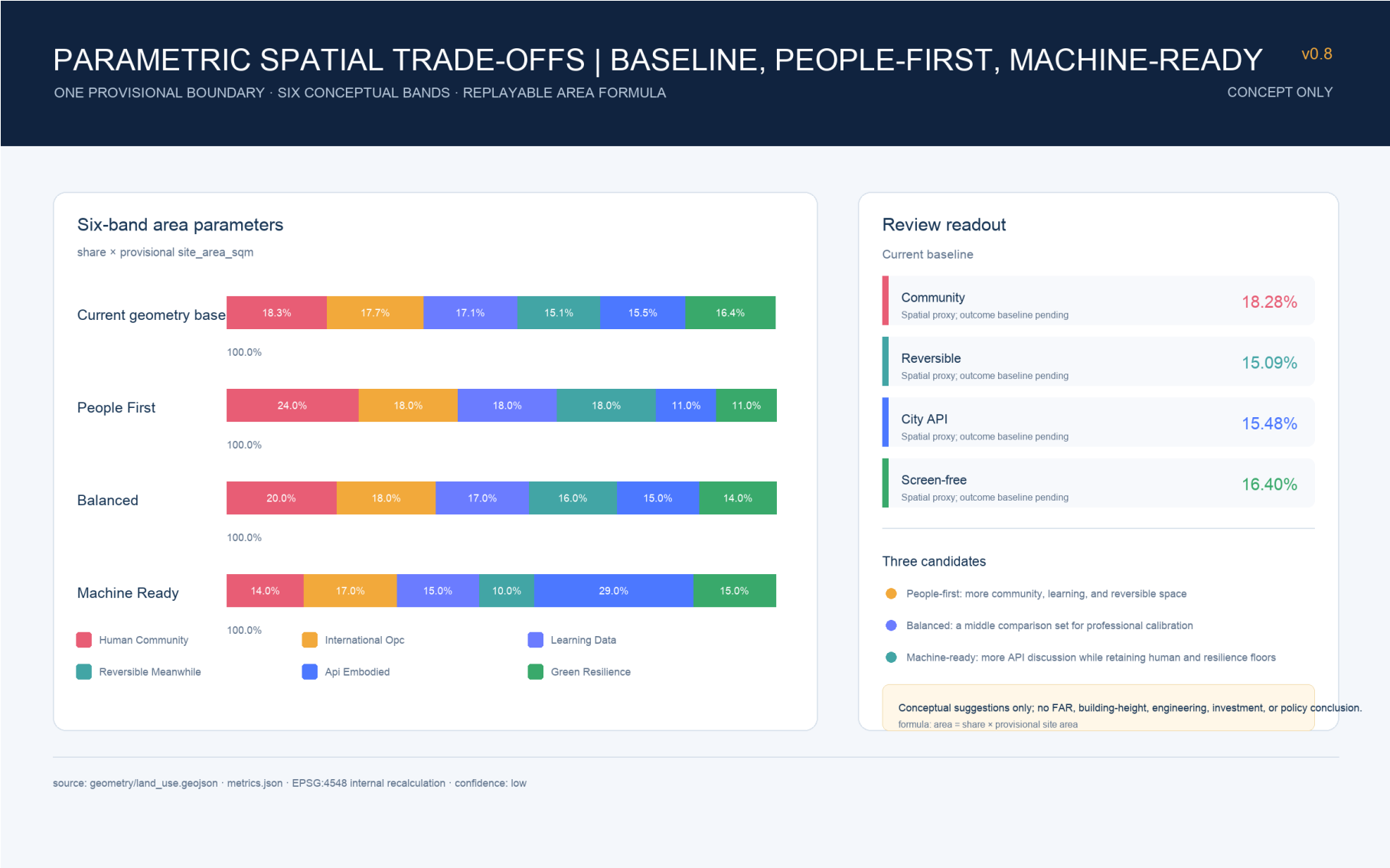
This iteration adds `HUMAN-CITY-RECEIPT-0.1`, a synthetic human-city handoff receipt. It records six H0-H5 gates for problem and boundary, data and choice, human accountability, bounded synthetic rehearsal, independent replay, and version exit. Three synthetic cases cover human channels and older residents, reskilling and workers at substitution risk, and the technology-service wing for small teams. [data:visual/assets/human-city-receipt.json] [data:visual/assets/human-city-receipt-evidence.json]

The network-free `run-human-city-receipt.js` checks 3/3 cases, 6/6 handoff gates, 6/6 spatial references, an equivalent human path, and no-write replay fields. Negative samples missing the human alternative or spatial anchor must fail. The receipt proves only that synthetic structure and stop conditions can be replayed; it is not evidence of real operations, staffing, permits, safety, performance, or public outcomes. [data:visual/assets/run-human-city-receipt.js] [depth:phasingimplementation]

v0.8: Make the technology-service wing and spatial trade-offs reviewable

This iteration places the taskbook’s Zhongguancun technology-service wing in `SC-D04`, the “factor configuration desk.” It links intellectual-property, compliance, data authorization, and cross-region application transfer for one-person companies, small teams, research teams, and Beijing-Tianjin-Hebei application partners. The card and space–operation matrix retain staffed advice, paper materials, authorization-chain checks, conflict disclosure, and a freeze condition. No institution, capital provider, or signed agreement is named. [source:AGENT-TASKBOOK] [data:geometry/constraints.geojson#ZONE-C-DATA] [data:geometry/publicspace.geojson#PUBLIC-D-GLOBAL] [depth:overallspatialstructure]

The parametric study takes the current six-band conceptual allocation as a baseline and compares three transparent parameter sets: people-first, balanced, and machine-ready. `visual/assets/run-parametric-tradeoff-study.js` checks share sums, area formulas, baseline reconnection, four objective lenses, comparison floors, and a non-dominance condition; formal `metrics.json`, geometry, drawings, and release state remain unchanged. Figure 11 makes the trade-off between community retention, learning, reversible space, city API, and resilience visible. Professional teams can replace the inputs once official boundaries, controls, rights, mobility, municipal, energy, and public baselines are available. [data:geometry/landuse.geojson#LU-H01] [metric:designpartitionareasqm] [depth:landuselayout] [depth:metricsrecalculation]



All three parameter sets remain low-confidence conceptual suggestions for professional teams. They compare spatial relationships only and generate no FAR, building-height, engineering, investment, employment, or policy conclusion. [data:geometry/siteboundary.geojson#SITE-001] [metric:communityretentionsupportarearatio] [metric:reversiblespaceratio]

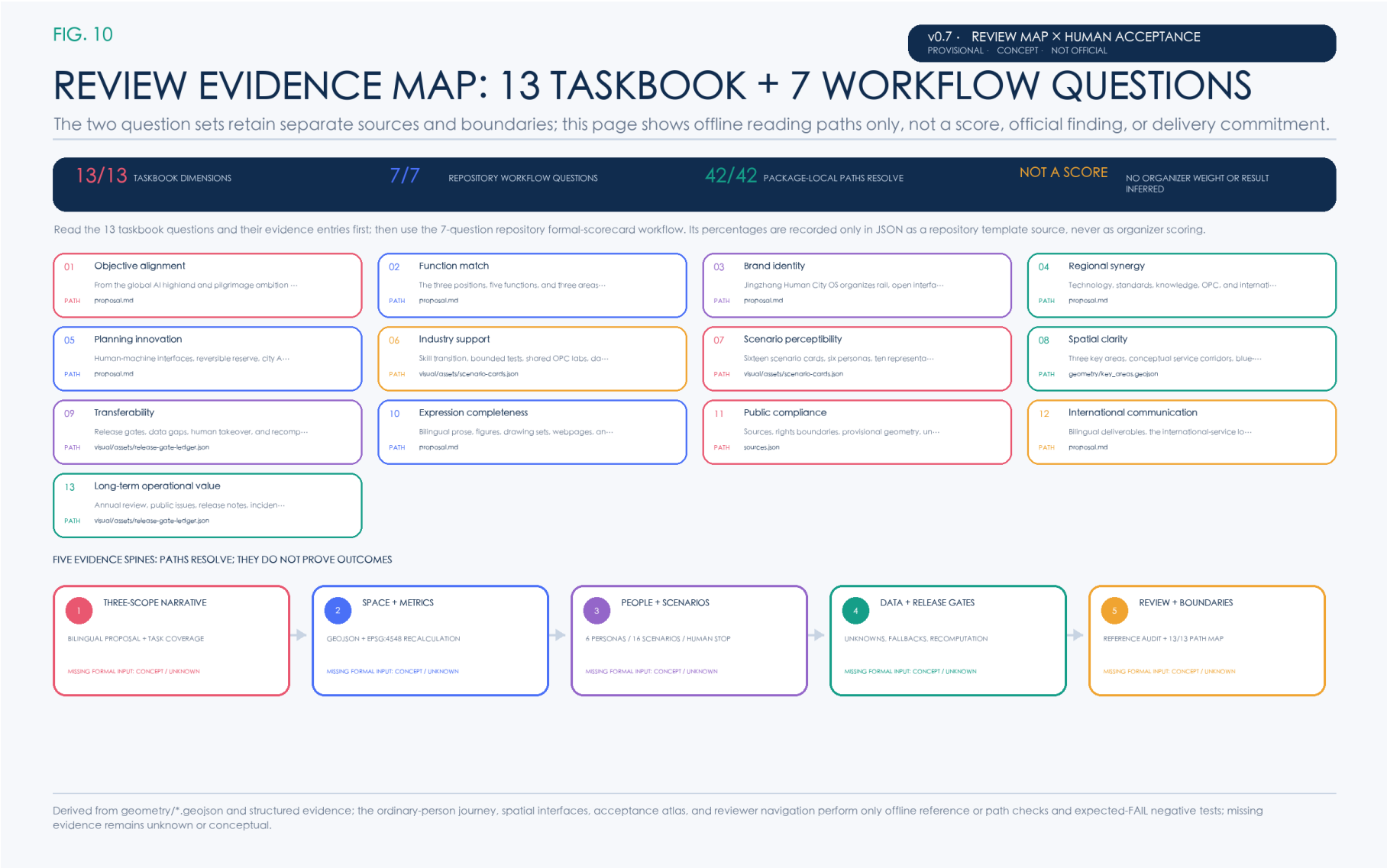
Jingzhang Human City OS

> From an AI showcase to a city for people in the AI era. AI is not the protagonist of the city; it is an auditable, rejectable, and reversible capability serving residents, workers, founders, and public institutions.

v0.7: Make Two Review Question Sets Return to the Same Evidence

This iteration repairs a readability gap, not a score opportunity. The repository formal scorecard has seven workflow questions with template percentages, while the taskbook lists thirteen unified review dimensions and registers no local weights. They must not be presented as one “official scoring” system. `reviewer-navigation-index.json` therefore records them separately: the first names only repository-script sources and workflow percentages; the second preserves every taskbook question without inferring a weight or total. Both paths return to the same bilingual prose, geometry, scenario, persona, release-gate, data, and rights ledgers. [source:AGENT-TASKBOOK] [depth:riskmissingdata]

The new network-free `run-reviewer-navigation-audit.js` reads the taskbook, local formal-scorecard scripts, and the package navigation. It verifies all 13/13 taskbook dimensions, 7/7 workflow questions, their 100% template-weight alignment, and 42/42 package-local file paths; negative samples that remove a dimension or replace a path with a nonexistent file must fail. It proves only that reading paths and local vocabularies have not drifted—not design quality, source authority, rights, field performance, reviewer judgement, or any score. Figure 10 compresses the two question sets, five evidence spines, and their shared boundary—missing input stays conceptual or `unknown`—onto one reading page. [data:geometry/siteboundary.geojson#SITE-001] [metric:siteareasqm] [depth:metricsrecalculation]



v0.8: Turn “ready to start” into a pre-authorization acceptance register

This iteration answers the most important delivery question in the review: the scenario cards already describe human takeover and stop actions, but they do not yet make clear when a study may start. `visual/assets/pilot-readiness-register.json` therefore registers all 6 `testvalidation=true` scenarios against a baseline, participant and time window, success condition, stop condition, human alternative, final accountability, review period, and deletion proof. Every value stays at “not collected,” “to be defined,” or “freeze before authorization”; no example number is presented as a field result. [data:geometry/phasing.geojson#PHASE-V02] [depth:riskmissingdata]

- | Question before entry | Current status | Who still has to complete it |
- | --- | --- | --- |
- | Baseline and observation scope | Not collected; sampling frame and time window are not frozen | An authorized party and professional evaluator after consent |
- | Success and stop conditions | Numeric thresholds freeze before authorization; scenario-card stops for safety, complaints, unauthorized access, accessibility barriers, and unclear accountability already apply | Professional evaluator and on-site safety role |
- | Human alternative and accountability | Staffed, telephone, paper, or on-site paths are registered; named shifts, response time, and operator remain unconfirmed | Operator, data-accountability party, and independent reviewer |
- | Review, objections, and deletion | Period, public summary, objection response, and deletion proof must enter the release note before start | Authorized party and public-feedback procedure |

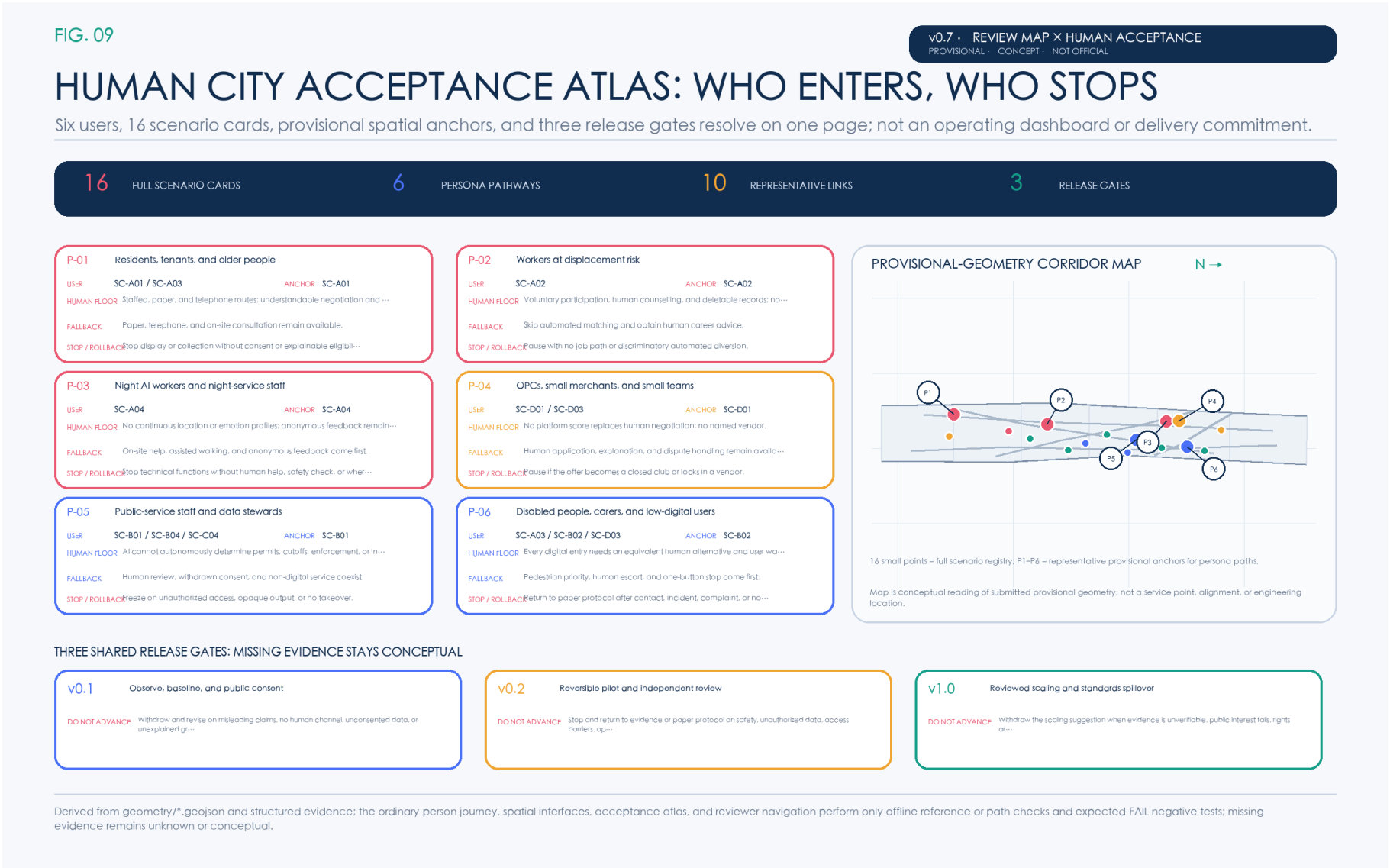
This is a pre-authorization working contract. The package contains no site visit, resident interview, survey, operating log, or real performance result; `sitevisitstatus`, `residentvalidationstatus`, `stakeholderengagementstatus`, and `fieldmeasurementstatus` remain not conducted or not collected. The offline runner checks 6/6 test scenarios, 8/8 required fields, and two negative fixtures for a missing field and an invented numeric value. A pass proves only that the register has no structural omission; it does not prove approval, deployment, or effectiveness. [metric:manualserviceequivalencerate] [depth:metricsrecalculation]

v0.6: Put “who can enter and who can stop” on one spatial page

This iteration reorganizes the acceptance relationships already distributed across six persona records, seventeen scenario cards, three release gates, and provisional spatial anchors into Figure 09. The six persona paths start with existing residents/older people, workers at displacement risk, night AI workers, OPCs/small merchants, public-service staff, and disabled or low-digital users. Ten directly related scenarios are selected as legible paths, while all seventeen cards, sixteen geometry nodes, and all three release gates remain visible so that representative personas stay alongside the complete scenario registry. [data:geometry/constraints.geojson#SC-A01] [metric:scenarionodecount] [depth:municipalnewinfrastructure]

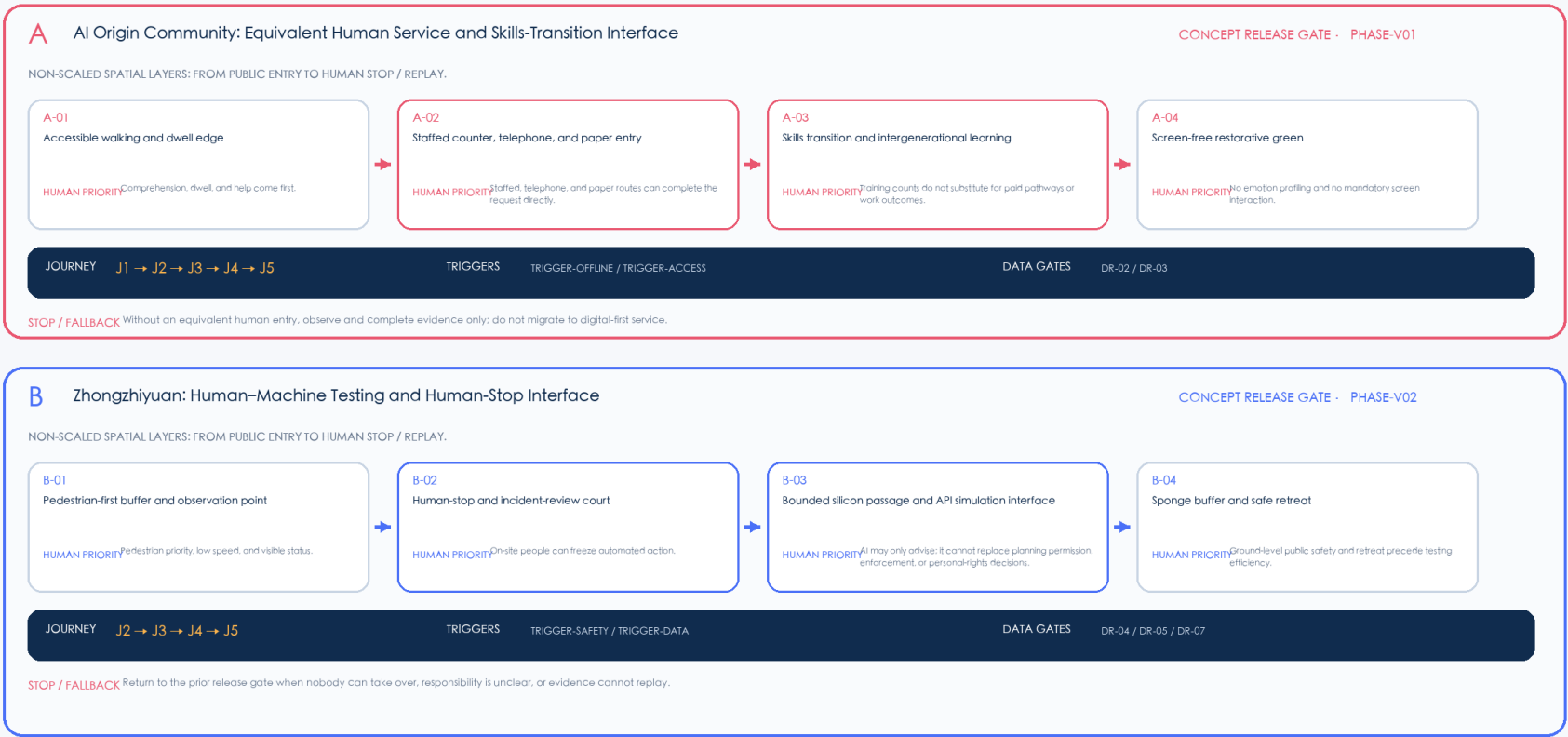
`visual/assets/human-city-acceptance-atlas.json` records only existing persona, scenario, and gate IDs. Its offline runner resolves 6/6 persona paths, 17/17 scenario cards, 10/10 representative links, 30/30 representative spatial references, 47/47 scenario spatial references, and 3/3 release gates; a fabricated persona anchor must fail. PASS proves only that the figure has not hidden the package's references and stop conditions. It does not prove real human coverage, staffed service, spatial access, field safety, permits, or social performance. [data:geometry/constraints.geojson#SC-C04] [data:geometry/phasing.geojson#PHASE-V10] [depth:riskmissingdata]

The six Figure 09 cards retain the same three questions—human floor, human fallback, and stop/rollback. For example, community retention cannot use a spatial proxy as a real retention rate; skills pathways cannot substitute training counts for employment outcomes; digital entry cannot displace an equivalent human alternative; and public-service AI cannot autonomously decide individual rights. The points remain conceptual anchors in submitted provisional geometry, not actual service points, road alignments, or engineering locations. [data:geometry/publicspace.geojson#PUBLIC-A-INCLUSION] [data:geometry/roads.geojson#ROAD-A-SKILL] [metric:manualserviceequivalencerate]



HUMAN-MACHINE INTERFACES: STOP ACTIONS

Non-scaled concepts: human priority and review precede testing. Not an engineering section or permit.



Derived from geometry/*.geojson and structured evidence; the ordinary-person journey, spatial interfaces, acceptance atlas, and reviewer navigation perform only offline reference or path checks and expected-FAIL negative tests; missing evidence remains unknown or conceptual.

v0.4: Ordinary-Person Service Journey and Data Readiness

First accept one service chain that an ordinary person can complete

From a reviewer's and an ordinary person's perspective, the minimum acceptance object of the Human City OS is not a city API. It is a service chain that a person can understand, refuse, and complete: enter a public space or service point → choose staffed, telephone, paper, or AI-assisted access → request one everyday service → let a person take over when the network is offline, accessibility is blocked, a safety event occurs, or data authorization is unclear → freeze, reroute, appeal, or exit → let an independent reviewer replay the evidence and decide whether to repair, continue, or withdraw. This remains a conceptual acceptance contract, not an operational, permit, or performance claim.

| Step | Space/service visible to an ordinary person | Evidence to replay | Fail-closed action |

| --- | --- | --- | --- |

- | 1. Enter and choose | Community service room, station/public space, staffed counter, telephone, paper, and AI-assisted entry shown together | Entry mode, service-user category, place/version, and accessibility note | Stop at observation and data completion when no equivalent human entry or clear data boundary exists |
- | 2. Request | A clear request for translation, queuing, accessible navigation, reskilling, or public-data advice | Purpose, minimum data, accountable role, time window, and alternative path | Register only, without invocation, when permissions exceed minimum, consent is missing, or accountability is unclear |
- | 3. Human takeover | A person takes over after outage, accessibility obstruction, safety event, or an unexplained AI recommendation | Trigger, takeover person, handoff time, state change, and complaint route | Freeze automated action and return to staffed/telephone/paper service; stop if nobody can take over |
- | 4. Exit and appeal | A person can cancel, reroute, delete data, object, and obtain a non-AI explanation | Exit reason, deletion/restoration action, alternative service, and unresolved item | Withdraw the scene and return to the prior release gate when rights are harmed, evidence is insufficient, or recovery is impossible |
- | 5. Independent replay | A reviewer replays one service chain and decides continue, repair, or withdraw | Minimal log, grouped result, objection handling, version, and review decision | Do not expand or report a display PASS as an outcome when the chain cannot be replayed or the slowest group worsens |

The contract is stored in `visual/assets/human-city-ordinary-journey.json`. It binds two minimum replay routes to existing scenario cards, personas, spatial anchors, and release gates: an accessibility/low-digital-capability route uses `SC-A03 + P-06 + PHASE-V01`, while a data-accountability route uses `SC-B01 + P-05 + PHASE-V02`. The offline `run-human-city-ordinary-journey.js` resolves those references and runs two negative samples: a missing human stop action and a nonexistent spatial anchor must both return `FAIL`; each of the five steps also binds trigger IDs and evidence fields, so the runner requires 5/5 steps, 4/4 triggers, and every registered evidence field to resolve, with unique IDs. This makes both reference resolution and rejection of a missing contract element reviewable; it remains contract-level negative testing, not field behavior or performance testing. It touches no personal data or external system; `performanceresults=null` and `operationalstatus=notauthorizednotrun`. A local PASS proves only that the contract and its negative tests can be replayed, not real accessibility, staffing, public acceptance, or safety performance. [data:geometry/phasing.geojson#PHASE-V02] [metric:versionreleasecount]

Reviewer navigation: separate the question sources, then replay the shortest evidence

The seven rows below correspond only to the repository formal-scorecard workflow. Their template percentages are not organizer scoring and are not a score for this package. The taskbook's thirteen unified review dimensions, their bilingual boundaries, and their shortest evidence paths are separately registered in `visual/assets/reviewer-navigation-index.json` and checked for coverage and paths by `run-reviewer-navigation-audit.js`. The package's `[metric:]`, `[source:]`, `[data:]`, `[depth:]`, and `[standard:]` markers remain resolved by the network-free `visual/assets/run-human-city-reference-audit.js`; an unregistered ID or nonexistent spatial feature is reported as FAIL.

| Repository formal workflow question (not organizer scoring) | Shortest entry | What to inspect first | What this package still cannot prove |

| --- | --- | --- | --- |

- | Brief alignment | `compliancematrix.json` + `designdepthmatrix.json` | Scope, task coverage, and deliverable depth | Provisional geometry is not a statutory redline or control plan |
- | Originality | `human-city-ordinary-journey.json` + `release-gate-ledger.json` | Whether human takeover, refusal, exit, and rollback enter the spatial-service chain | An offline contract is not field performance or deployed capability |
- | AI planning innovation | `run-human-city-ordinary-journey.js` + `parametric-tradeoff-study.json` + `scenario-cards.json` | Whether planning alternatives, AI-assisted reasoning, and human takeover are recalculable and reviewable | Offline runners do not replace professional models, permits, or field tests |
- | Implementation feasibility | `implementation-operation-matrix.json` + `data-readiness-register.json` | Accountability, prerequisites, release gates, and stop conditions | Suggested roles are not confirmed operators; budget, approvals, and field baselines remain open |

| Public interest and inclusion | `personas-and-fairness.json` + `accessibility-audit.json` | Equivalent access, refusal, human fallback, and appeal paths | Real group experience, coverage, consent, and outcomes require walk-throughs and sampling |

| Risk and compliance | `assumptions.json` + `rights-clearance-ledger.json` | Sources, authorization, rights, provisional geometry, and pause conditions | Registration is not a permit, rights clearance, or safety review |

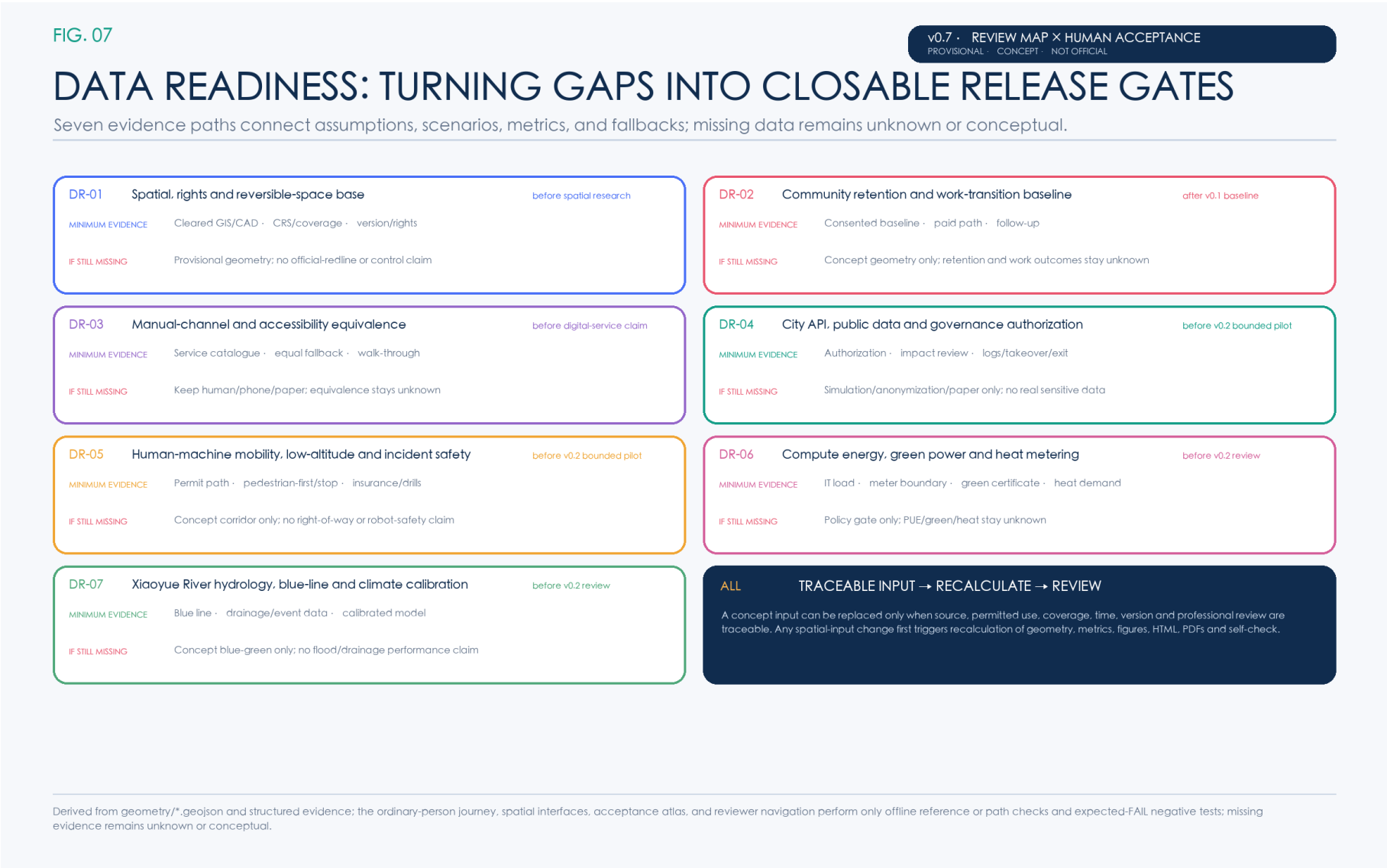
| Expression completeness | Bilingual proposal + `visual/index.html` + `manifest.json` | Whether text, drawings, visuals, and ledgers return to the same evidence | Navigation completeness does not raise the evidence level |

Data readiness: turn evidence gaps into closable release gates

v0.4 is also a substantive upgrade to data governance and social-outcome boundaries, not an operation on date, name, slug, or public ordering. The new `visual/assets/data-readiness-register.json` uses seven evidence paths to connect twelve assumptions to scenarios, affected spatial layers, metrics, release gates, minimum evidence, suggested review roles, fallbacks, and recalculation triggers. It appoints no real operator, approval date, or implementation commitment. Its purpose is to make an evidence gap traceable and vetoable while keeping an insufficiently evidenced item explicitly conceptual. [data:geometry/phasing.geojson#PHASE-V02] [depth:riskmissingdata]

This iteration also records `sustainedemploymenttransitionrate` and `manualseviceequivalencerate` as formal `unknown` metrics. The first requires a consented baseline, a paid-pathway definition, and predeclared follow-up; the second requires a service catalogue, equal-status fallback rules, and real walk-through evidence. Corridor length, training attendance, web accessibility, or global AI-exposure rates cannot substitute for either outcome. No provisional-boundary coordinate, design partition, existing area, ratio, line length, or existing metric value changes in this iteration. [metric:sustainedemploymenttransitionrate] [metric:manualseviceequivalencerate] [metric:skilltransitioncorridorlengthm]

Figure 07 compresses each path into a review card of “minimum evidence—entry gate—if still missing,” while the full JSON retains the assumption, scenario, spatial, metric, and fallback relationships. When any official or cleared spatial input arrives, geometry, metrics, figures, HTML, PDFs, and self-check must be recalculated before a claim changes; a paragraph or partial sketch cannot substitute. [metric:siteareasm] [depth:metricsrecalculation]



v0.3: Making Spatial Evidence Genuinely Legible

v0.3 is a substantive upgrade to spatial-evidence presentation, not an operation on date, name, or public ordering. Without changing a single provisional-boundary coordinate, design partition, area, ratio, line length, scenario count, or `unknown` state, it redraws six bilingual figures and bilingual A3/A0 drawing sets. The figures use the same package GeoJSON and internal EPSG:4548 recalculation, add figure numbers, scale bars, corresponding legends, and provisional-boundary notes, and apply a 90-degree clockwise display rotation solely to make the corridor readable; original north is therefore explicitly labelled “N →”. This is a display transformation only, never an upgrade of provisional geometry to an official redline or survey product. [data:geometry/siteboundary.geojson#SITE-001] [metric:siteareasm] [depth:metricsrecalculation]

Every layer continues to derive directly from `geometry/` and `metrics.json`. The six land-use polygons in Figure 02 still cover the submitted boundary seamlessly and without overlap, while the three key areas in Figure 03 and service corridors in Figure 04 remain conceptual suggestions only. `visual/assets/spatial-figure-method.json` records inputs, display rotation, and precision limits so professional teams can regenerate the entire package after official polygons, controls, rights, roads, utilities, blue lines, heritage, existing-condition, and operational data arrive; prose or visual completeness may not substitute for those data. [data:geometry/landuse.geojson#LU-H01] [metric:designpartitionareasm] [depth:landuselayout]

v0.2: Turning City Versions into Vetoable Release Gates

This v0.2 is a substantive evidence upgrade, not a repackaging of spatial values or public ordering. Without changing provisional-boundary coordinates, design areas, or existing `known` metrics, it adds seventeen cards linking space, accountability, human fallback, data boundary, and stop condition, a ledger for three city release gates, and audits for accessibility and persona fairness. The earlier claim of reversibility becomes reviewable questions of who may begin research, what evidence is needed, and when a proposal must pause. It remains a conceptual reference scheme for professional teams to deepen. [data:geometry/phasing.geojson#PHASE-V02] [metric:versionreleasecount]

| Release gate | Must be completed before entry | Reviewable evidence | Pause or rollback condition |

| --- | --- | --- | --- |

| v0.1 Observe, baseline, and public consent | Data boundary, human-service floor, objection channel | Comprehension walk-through; problem and gap ledger | Unconsented data use, no human channel, or exclusion risk |

| v0.2 Reversible pilots and independent evaluation | Limited place/time/data/accountability, human takeover, professional authorization | Incident review; authorization, metering, calibration, and independent assessment | Safety event, unauthorized data, accessibility barrier, or insufficient evidence |

| v1.0 Reviewed scaling and standards spillover | Prior-version evidence, objection handling, public-interest review | Annual health check, release note, reviewable protocol drafts | Unverifiable evidence, harmed rights, or changed base conditions |

The gate ledger and scenario cards expressly do not constitute permits, engineering schemes, investment commitments, or implementation findings. Once official boundaries, rights, controls, roads, utilities, blue lines, or existing-condition data arrive, the whole package must be recalculated before another version is considered.

Core Concept and Four Operating Mechanisms

The core concept is the Human City OS: a release system that organizes urban space, public services, and AI agents so they remain observable, rejectable, and reversible. AI is not turned into a landmark, and automation does not replace human judgement. Each urban capability is split into four auditable mechanisms: a human-priority mechanism that protects staffed service, accessibility, and the right to refuse; an evidence-release mechanism that binds place, time, data, accountability, and version gates; a space-service interface mechanism that connects the city API, walking network, public space, and scenario cards into a reviewable service chain; and an exit-and-repair mechanism that requires shutdown, removal, data deletion, objection handling, and independent review together. [depth:phasingimplementation] [metric:versionreleasecount]

Together these mechanisms turn “from an AI showcase to a city for people” from a value statement into a spatial-governance-operations protocol. Every pilot must answer who may start it, who retains final responsibility, how the public can refuse, what evidence permits expansion, and how the system recovers after failure. Unknown metrics remain `unknown` rather than being filled by model inference or case analogy; every spatial band remains a design suggestion, not a statutory land-use zone, official boundary, or implementation commitment.

Design Basis and Source List

The proposal first follows the three planning extents, three key areas, and design tasks established by the official call, and uses the six agent tasks in the cleared taskbook as its delivery checklist. [source:OFFICIAL-ANNOUNCEMENT] [source:AGENT-TASKBOOK] The five formally usable records in the repository source registry support only the uses explicitly permitted there. Provisional boundaries are used for generation, visualization, and intake checks; they are not represented as statutory redlines, official parcels, or exact regulatory controls. [source:SOURCE-REGISTRY] [data:geometry/siteboundary.geojson#SITE-001]

The available public package does not contain official polygons, plot ratios, height or density controls, green-space ratios, setbacks, road redlines, property rights, existing-building surveys, heritage control lines, utility capacity, or a public-service baseline. Land-use bands, concept buildings, roads, a low-altitude layer, sponge facilities, and phases are therefore design hypotheses requiring professional development. Every material limitation is registered in `assumptions.json`. The recalculated area of 11,412,825.386 square metres is an internal consistency value derived from the provisional geometry, not an official area conclusion. [metric:siteareasqm] [depth:existingconditionsdiagnosis]

New claims about employment, talent, and energy use verifiable first-party sources. The IMF reports that almost 40% of global employment is exposed to AI and that the share is about 60% in advanced economies; exposure includes both substitution and augmentation and is not a forecast of unemployment. In the World Economic Forum's 2025 employer survey, 41% of respondents expected workforce reductions where AI can replicate roles; this is not a Haidian forecast. [source:IMF-AI-JOBS-2024] [source:WEF-FUTURE-JOBS-2025] The currently verifiable public figure for Haidian is approximately 90,000 AI professionals, not 95,000. It is used only to establish the order of magnitude relevant to wellbeing and night services, never to infer a site population or facility capacity. [source:HD-AI-TALENT-2026]

Compute-energy requirements are also corrected to current official wording. Beijing's 2024 computing-infrastructure implementation plan states that PUE for new or expanded intelligent-computing centres should generally not exceed 1.25, and generally not exceed 1.15 for large advanced facilities. From 2026, data centres exceeding the applicable 1.35 limit are subject to differentiated electricity pricing. A 30% green-electricity share is a Beijing target for a specified band of existing data centres; this proposal may use it as a design baseline, but does not misstate it as a statutory minimum for a new centre. [source:BJ-COMPUTE-2024] [source:BJ-DATACENTER-2024]

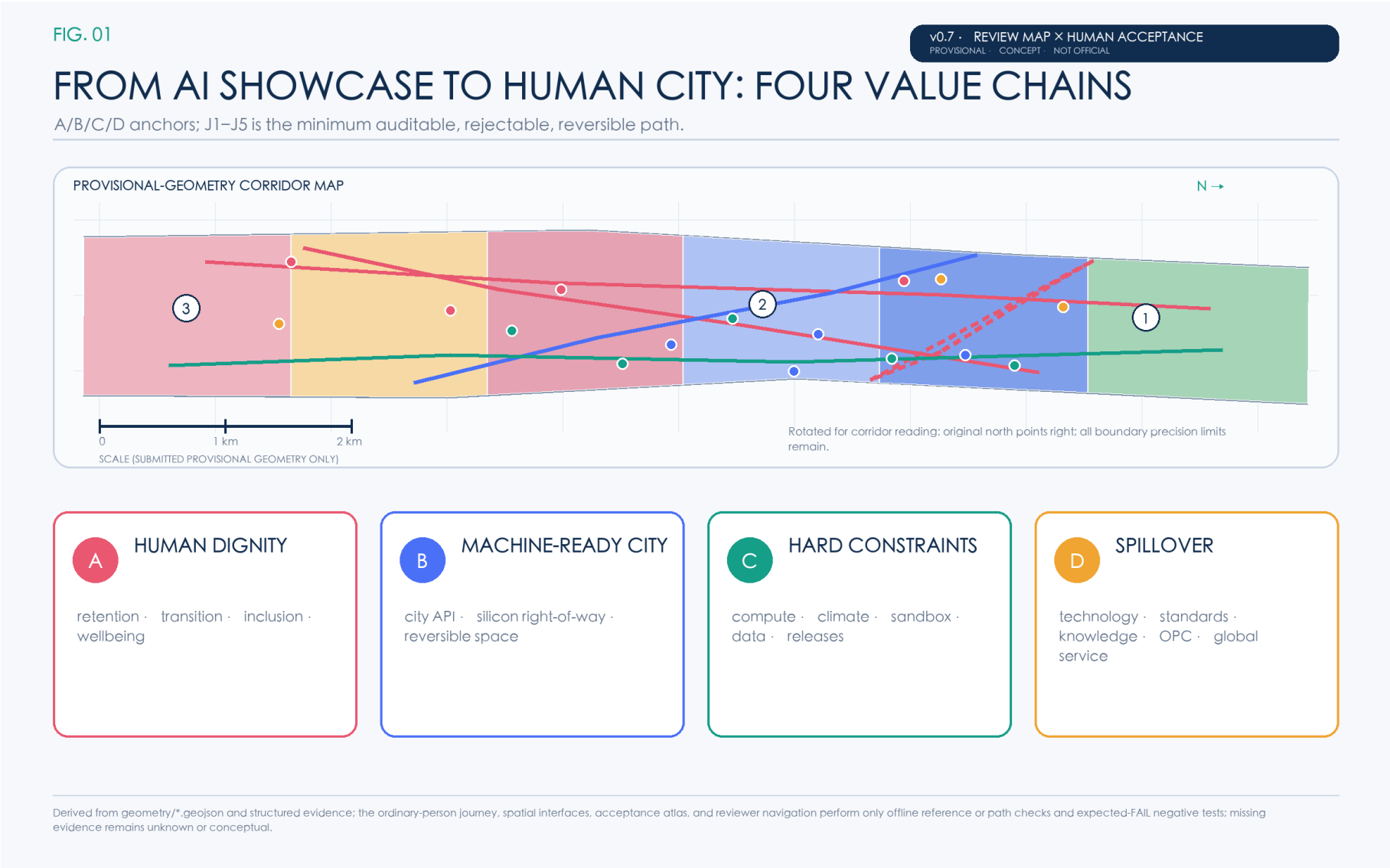


Figure 01 key-area legend: 1 = Zhongzhiyuan; 2 = Beijing AI Origin Community; 3 = Dazhongsi AI Industry Cluster. All three are provisional key areas for conceptual validation and professional deepening, not official redlines.

Three-Level Scope Framework

The 43.6-square-kilometre coordinated research area asks how Haidian's innovation capacity can spill over beyond the district. The approximately 11.4-square-kilometre overall design area asks how human services, a city API, and hard constraints can coexist spatially. The approximately 368.4-hectare key-area set asks what should be tested first in Zhongzhiyuan, the AI Origin Community, and Dazhongsi. The three extents form one evidence chain running from strategy through spatial structure to controlled validation.

[standard:PROJECT-OFFICIAL-ANNOUNCEMENT] [depth:threelevelscopeframework]

| Extent | Primary question | Delivery in this proposal | Boundary condition |

| --- | --- | --- | --- |

| Coordinated research area | How can technology, standards, and knowledge spill over? | Six global cases, Beijing-Tianjin-Hebei collaboration, OPC support, and international services | No fabricated firms, investment, output, or partnerships |

| Overall design area | How can an AI showcase become a city for people? | Four value chains A/B/C/D, six spatial layers, and paired human/machine movement networks | Functional bands are not statutory land-use zones or road redlines |

| Three key areas | How can reversible pilots be validated? | Sixteen geometry nodes, seventeen scenario cards, six test-and-validation scenes, and three release stages | No operation without authorization; heritage, traffic, airspace, and data rules remain controlling |

Group A, Human Dignity, covers community retention, work transition, digital inclusion, and physical and mental wellbeing. Group B, Machine-Ready City, covers a city API, silicon right-of-way, a digital twin, and reversible white space. Group C treats compute-energy, climate, a regulatory sandbox, public-data authorization, and city-version governance as hard gates. Group D organizes technology, standards, knowledge, OPC support, and international services into spillovers. Each group appears in scenario nodes, spatial layers, and recalculable metrics rather than remaining a slogan. [data:geometry/constraints.geojson#SC-A01] [metric:scenariodecount]

The spatial proposition is “one historic public spine, two movement systems, three validation grounds, four value chains, and six functional bands.” The Jingzhang Railway Heritage Park is treated only as a narrative spine of history and public space; the provisional boundaries remain low-contrast dashed references. Any official boundary update triggers a full regeneration of every design layer, figure, HTML page, PDF, and metric instead of a selective graphical replacement. [data:geometry/phasing.geojson#PHASE-V01] [depth:overallspatialstructure]

Coordinated Research Area: Industry and Future City Research

The ecosystem is not reduced to attracting large firms. It is decomposed into six capabilities: frontier research, affordable testing, trustworthy data, workforce transition, social acceptance, and international services. Large teams can provide foundation models and platforms. Small teams and one-person companies can enter public-interest scenarios through shared laboratories, model and compute services, legal support, and data-authorization assistance. Workers whose occupations are changing can move through a reskilling corridor toward robotics operations, data quality, scenario operations, accessibility services, and incident review. [source:HD-OPC-2026] [depth:overallspatialstructure]

Six international cases contribute mechanisms, not imported urban forms. Singapore’s Punggol Digital District demonstrates how an open digital platform and digital twin can become callable district infrastructure. Helsinki’s Smart Kalasatama demonstrates the value of an inhabited community, resident participation, and agile pilots over a one-time “smart city delivery.” Barcelona’s 22@ experience warns that innovation-district renewal must address housing and existing communities. Seoul AI Hub combines education, incubation, research, and public support. Busan Eco Delta City places living labs, a regulatory sandbox, open data, and robotics within the same public program. Toronto’s Quayside debate shows why independent governance, responsible data use, and public purpose must precede sensor deployment. [source:CASE-PUNGGOL] [source:CASE-KALASATAMA] [source:CASE-QUAYSIDE]

| Case | Transferable mechanism | Jingzhang translation | Required safeguard |

| --- | --- | --- | --- |

| Punggol Digital District | Open platform, APIs, digital twin | City API Exchange | Data catalogue, permissions, logs, and exit procedure |

| Smart Kalasatama | Resident participation and agile pilots | Reversible v0.1-v0.2 pilots | Start with public problems, not technology in search of a use |

| Barcelona 22@ | Innovation alongside housing and community | Community-retention ledger and small-business return | Baseline, eligibility, leases, and anti-displacement audit |

| Seoul AI Hub | Education, incubation, and R&D support | Reskilling corridor and shared lab | Track employment outcomes, not training attendance publicity |

| Busan Eco Delta | Living lab, sandbox, open data | Embodied-AI testing and climate simulation in the Xiaoyue River wing | Insurance, incident response, and human takeover |

| Toronto Quayside | Independent data governance | Public Data Authorization Room | Public interest, privacy, and third-party evaluation first |

Spillover follows four paths: technology, standards, knowledge, and international services. Technology spillover gives Beijing-Tianjin-Hebei manufacturing and application partners test interfaces for Haidian research. Standards spillover converts right-of-way, authorization, accessibility, and incident-review protocols into reusable templates. Knowledge spillover combines a fifteen-minute encounter network, an open problem list, annual release notes, and cross-campus communities into traceable knowledge assets. International services cover language, medical navigation, intellectual property, compliance, and daily life; the proposal fabricates no school, hospital, agency, or institutional commitment. The Zhongguancun technology-service wing supplies a public catalogue, staffed advice, authorization review, and an application-transfer ledger for this spillover. [source:AGENT-TASKBOOK] [data:geometry/constraints.geojson#ZONE-C-DATA] [depth:existingconditionsdiagnosis]

The Zhongguancun technology-service wing is a conceptual interface for global factor configuration, Zhongguancun intellectual-property support, and compliant capital-information services. `SC-D04` sends reviewable protocol drafts into Beijing-Tianjin-Hebei manufacturing and application discussions; it does not promise financing, investment, recruitment, or a confirmed operator. The Xiaoyue River scenario-empowerment wing carries embodied testing and climate-resilience scenes, so the two wings connect through standards, data, knowledge, and workforce transition. [source:AGENT-TASKBOOK] [data:geometry/publicspace.geojson#PUBLIC-D-GLOBAL] [depth:overallspatialstructure]

Overall Design Area: Urban Renewal and Regulatory-Plan-Level Urban Design

The overall design establishes six functional operating layers within the provisional design boundary while retaining the city’s diverse everyday structure: community retention and support; international and OPC services; reskilling and public-data learning; reversible white space; city API and embodied-AI R&D; and screen-free green/climate resilience. The six polygons cover the submitted site without overlap, and their areas and shares are recalculated from one GeoJSON source in EPSG:4548. [data:geometry/landuse.geojson#LU-H01] [metric:designpartitionareasqm] [depth:landuselayout]

The community-retention support band occupies approximately 18.3% of the provisional site, while reversible white space occupies approximately 15.1%. These are spatial proxies describing the design allocation; they are not a resident-retention rate or a statutory reserve ratio. A real resident-retention rate requires a rights-cleared household baseline, eligibility rules, informed consent, and longitudinal registration, so the outcome metric remains `unknown`. [metric:communityretentionsupportareatio] [metric:reversiblespaceratio] [metric:residentretentionrate]

The city API uses a six-step protocol: catalogue, authorize, invoke, log, audit, and exit. A municipal-facility agent may call only minimized and tiered interfaces. Water, electricity, heat, transport, and public-data calls retain human takeover and incident traceability. Beijing’s current public-data authorization rules emphasize legal compliance, public-interest priority, security, evaluation, and exit. The proposal therefore treats a “data-factor district” as an auditable institutional prototype, not a technology showroom. [source:BJ-PUBLIC-DATA-2026] [data:geometry/constraints.geojson#ZONE-B-API]

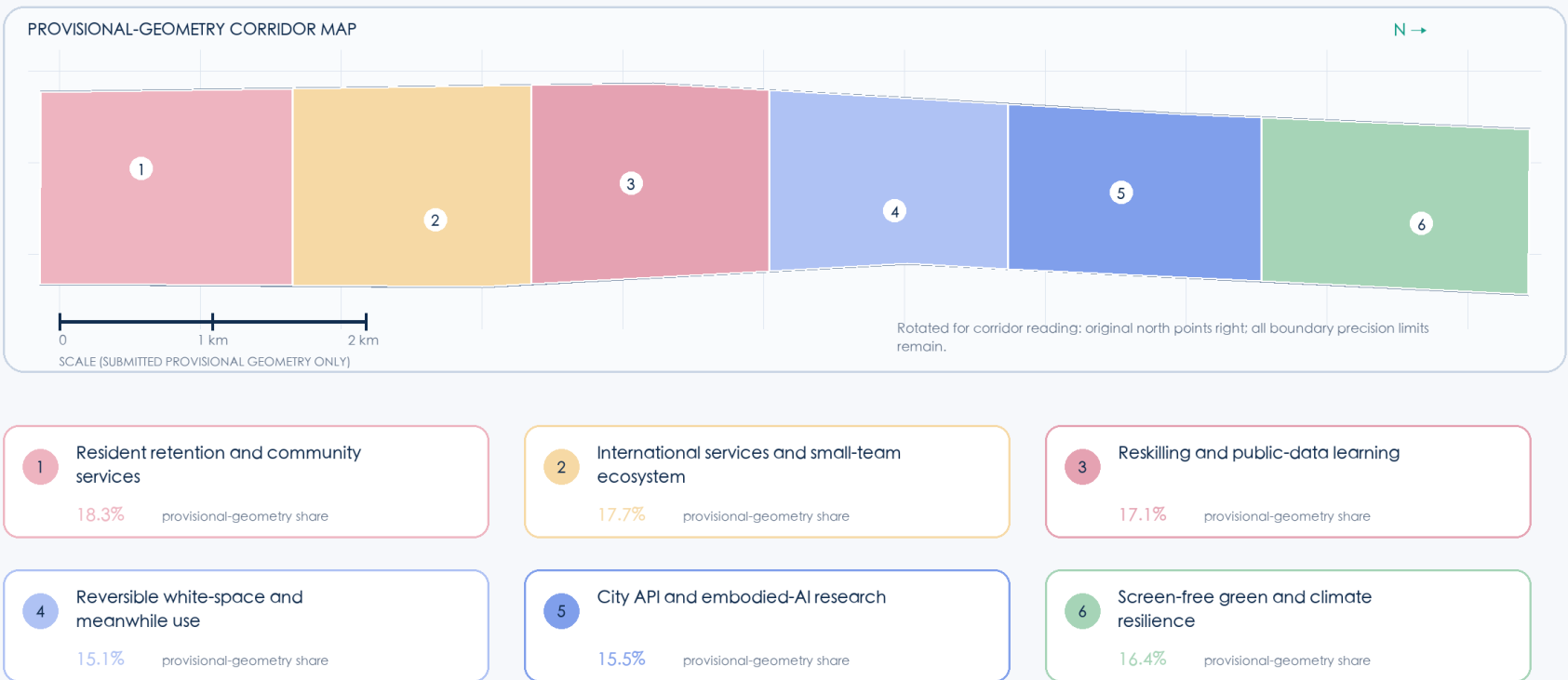
Reversible design uses modular components, removable service pods, short-cycle leases, and meanwhile use to mediate the mismatch between three-month AI cycles and ten-year urban construction cycles. Every pilot must file removal, restoration, and data-deletion plans at approval. A v0.2 pilot cannot expand to v1.0 without independent evaluation and public review. Building height, development intensity, construction volume, and retain-renovate-demolish decisions remain for professional development because official controls and surveys are absent. [standard:MOHURD-CONTROL-DETAILED-PLANNING] [depth:developmentintensitycontrols]

FIG. 02

v0.7 · REVIEW MAP X HUMAN ACCEPTANCE
PROVISIONAL · CONCEPT · NOT OFFICIAL

SIX-LAYER SPATIAL OS: PROVISIONAL PARTITIONS AND REVERSIBLE SPACE

Concept layers retain human entry and evidence gates; not zoning, FAR, height, or demolition decisions.



Derived from geometry/*.geojson and structured evidence; the ordinary-person journey, spatial interfaces, acceptance atlas, and reviewer navigation perform only offline reference or path checks and expected-FAIL negative tests; missing evidence remains unknown or conceptual.

Parametric study: a replayable candidate set for spatial trade-offs

`visual/assets/parametric-tradeoff-study.json` uses the current six-band geometry allocation as a baseline and compares people-first, balanced, and machine-ready parameter sets. Each area is computed as `share × provisional siteareasqm`; the offline runner checks share sums, derived areas, and baseline reconnection to the current metrics, with evidence written to `parametric-tradeoff-study-evidence.json`. The study puts community retention, learning, reversible space, city API, and screen-free resilience on one comparison page for professional review after formal inputs arrive. [data:geometry/landuse.geojson#LU-H01] [metric:communityretentionsupportarearatio] [metric:reversiblespaceratio] [depth:metricsrecalculation]

The three sets remain low-confidence conceptual suggestions. They do not change formal geometry, areas, ratios, drawings, or `scenarionodecount=16`; `SC-D04` reuses existing spatial anchors while the scenario registry contains seventeen cards. Any next-round choice must recheck official boundaries, rights, controls, buildings, mobility, utilities, energy, climate, and public baselines. [data:geometry/siteboundary.geojson#SITE-001] [metric:scenarionodecount] [depth:riskmissingdata]

PARAMETRIC SPATIAL TRADE-OFFS | BASELINE, PEOPLE-FIRST, MACHINE-READY

v0.8

ONE PROVISIONAL BOUNDARY · SIX CONCEPTUAL BANDS · REPLAYABLE AREA FORMULA

CONCEPT ONLY



source: geometry/land_use.geojson · metrics.json · EPSG:4548 internal recalculation · confidence: low

Detailed Design of Key Areas

Zhongzhiyuan, the AI Origin Community, and Dazhongsi use the repository's coarse provisional extents only to locate design questions; the rectangular outlines carry no planning meaning. All three areas follow the sequence “public problem, controlled test, independent evaluation, reversible expansion,” while each has a different first responsibility. [data:geometry/keyareas.geojson#PROV-KEY-001] [metric:keyareacount] [depth:threekeyareadetaileddesign]

| Key area | Primary proposition | First spatial actions | Evaluation gate |

| --- | --- | --- | --- |

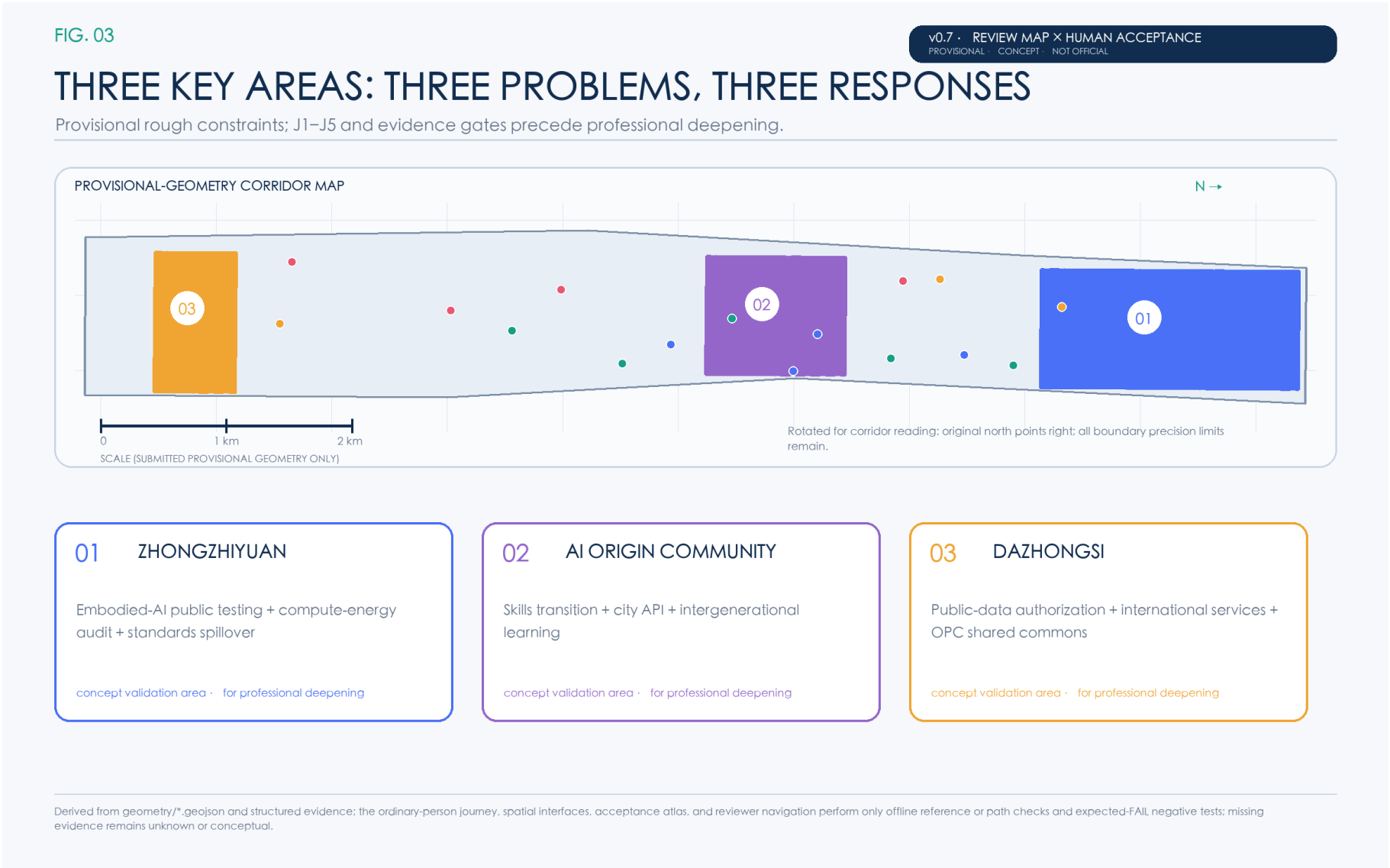
| Zhongzhiyuan | Can a machine-ready city enter public space safely? | Silicon-right-of-way test, embodied-AI public test bay, compute-energy/heat audit, standards spillover studio | Accountable operator, insurance, human takeover, energy metering, and public incident review |

| AI Origin Community | Can innovation density protect everyday life and workforce transition? | Reskilling Relay, City API Exchange, intergenerational learning, City Release Room | Employment conversion, retained human services, data minimization, and resident vote |

| Dazhongsi | Can data and international services create a trustworthy urban interface? | Public Data Authorization Room, International Service Commons, OPC shared services | Lawful authorization, exit, accessibility, multilingual and compliance-service audit |

Embodied-AI testing in Zhongzhiyuan does not assume that autonomous vehicles or robots naturally possess right-of-way. Temporary permits are conditioned on pedestrian priority, speed, time, task, and risk class. The low-altitude logistics layer is only a vertically separated concept corridor and cannot enter testing before airspace, noise, drop risk, privacy, fire safety, and insurance reviews. [data:geometry/roads.geojson#ROAD-B-SILICON] [metric:siliconrightofwaylengthm]

The AI Origin Community places workers at substitution risk, not only celebrated technical talent, inside the innovation chain. Capability assessment, paid learning, real-job trials, and roles in robotics operations, data quality, scenario operations, accessibility, and human review form a trackable reskilling corridor. Dazhongsi lowers compliance costs for small teams through public-data and international-service support. Any subsidy or company service is referenced to current policy and is not extended into a promise. [data:geometry/roads.geojson#ROAD-A-SKILL] [metric:skilltransitioncorridorlengthm]



AI Innovation Ecosystem, Personas, and AI+ Scenarios

Six personas establish that the right to refuse AI is part of service quality. Long-term residents and older people need a human channel, familiar social networks, privacy, and low-disruption renewal. Workers at substitution risk need a verifiable route to a new job. AI researchers need safe night travel, twenty-four-hour basics, and screen-free recovery. OPCs and small teams need on-demand labs, compute, and compliance support. Public-service staff need explainable work orders and human review. International visitors need language, medical navigation, intellectual-property, compliance, and daily-life support. [source:HD-AI-TALENT-2026] [depth:municipalnewinfrastructure]

| Persona | Non-negotiable right | Spatial/service response | Validation |

| --- | --- | --- | --- |

| Existing resident or older person | Ability to remain, human service, privacy, familiar relations | Community Retention Hall, staffed counter, intergenerational learning | Retention awaits a lawful baseline; audit human-channel availability |

| Worker at substitution risk | Paid transition period and a real job pathway | Reskilling corridor and job-trial workshop | Longitudinal tracking from enrolment to trial and sustained employment |

| AI professional | Safe return at night, health, non-screen rest | Night Safety Link, 24-hour basics, screen-free green space | Night-service coverage and anonymous satisfaction |

| OPC or small team | Low-threshold facilities and fair access to tests | Shared lab, model/compute, legal and data support | Open application, conflict disclosure, and resource audit |

| Public-service worker | Human final responsibility and explainable tools | City API console and Incident Review Court | Takeover drill and log audit |

| International visitor or team | Accessible language, health, compliance, and daily services | One-stop International Service Commons | Multilingual accessibility walk-through; no invented providers |

All seventeen scenario cards are encoded in the structured registry and anchored to sixteen geometry nodes: four geometry nodes in A, four in B, five in C, and three in D. Six are explicitly tagged as test-and-validation scenarios, exceeding the taskbook minimum. [data:geometry/constraints.geojson#SC-B01] [metric:humanfirstscenariocount] [metric:machinecityscenariocount]

v0.2 upgrades these sixteen entries from a catalogue to reviewable scenario cards. Each links its spatial anchor, suggested users, human alternative, data boundary, evidence required for acceptance, and stop condition. Test scenarios additionally use the four limits of place, time, data, and accountability. This does not mean that any scenario is ready to

operate; it makes what is still missing visible before expansion. [data:geometry/constraints.geojson#SC-C03] [metric:testvalidationscenariocount]

| Card | Group | Scenario | Spatial carrier | Human/institutional safeguard |

| --- | --- | --- | --- | --- |

| SC-A01 | A | Community Retention Ledger | Community-retention support area | Rights-cleared baseline, resident consent, no commercial profiling |

| SC-A02 | A | Reskilling Relay | Reskilling corridor | Paid learning and job-outcome tracking |

| SC-A03 | A | Human Channel and Intergenerational Learning | Public-service hall | Permanent staffed option; no mandatory QR access |

| SC-A04 | A | 24-hour Wellbeing Refill | Night link and screen-free gardens | Anonymous use; no emotion profiling |

| SC-B01 | B/test | City API Exchange | API authorization zone | Least privilege, logs, human takeover, and exit |

| SC-B02 | B/test | Silicon Right-of-Way Test Ground | Controlled Xiaoyue River route | Time/speed limits, insurance, and incident response |

| SC-B03 | B | Reversible Component Library | White-space/meanwhile-use band | Removal and restoration plan submitted at approval |

| SC-B04 | B/test | Public Digital-Twin Bench | City API zone | Synthetic or anonymized data first; compare model and reality |

| SC-C01 | C/test | Compute-Energy and Recovered-Heat Audit | Concept building node | Meter PUE, green power, and recovered heat |

| SC-C02 | C/test | Flood and Sponge Simulation Ground | Xiaoyue River resilience wing | Obtain hydrology, blue-line, and drainage evidence first |

| SC-C03 | C/test | Regulatory Sandbox and Incident Review Court | Public governance space | Responsibility, insurance, shutdown, and public review |

| SC-C04 | C | Public Data Authorization Room | Dazhongsi data-service zone | Public interest, privacy, and third-party evaluation |

| SC-C05 | C | City Release Room | Origin Community public hall | Annual health check, public issues, and release note |

| SC-D01 | D | OPC Shared Laboratory | Shared experiment/service node | Open application and supplier neutrality |

| SC-D02 | D | Standards Spillover Studio | Zhongzhiyuan collaboration node | Publish reviewable protocols; do not claim national-standard status |

| SC-D03 | D | International Service Commons | Dazhongsi public space | Multilingual, accessible, and compliance-bound services |

The six tests share a “limited place, limited time, limited data, limited responsibility” protocol. Success means jointly meeting safety, human-takeover, energy, accessibility, public-acceptance, and removability thresholds. A strong single technical indicator never automatically authorizes expansion. [metric:testvalidationscenariocount] [depth:riskmissingdata]

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

The six functional bands use a subset of the national land-use classification for machine readability, while remaining design suggestions rather than approved land uses. Community retention uses 0702; international and small-team services use 05; reskilling uses 0804; reversible white space uses 16; city API and R&D use 0802; and screen-free resilience space uses 1401. The six faces share edges, cover the provisional ‘SITEBOUNDARY’, and contain neither gaps nor overlaps. [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE] [data:geometry/landuse.geojson#LU-B01]

Eight building footprints locate program prototypes only: Community Retention Service Station, Reskilling Workshop, Age-Friendly Human Channel Station, City API Exchange, Embodied-AI Test Bay, Compute-Energy and Recovered-Heat Audit Station, OPC Shared Laboratory, and International Service Commons. Their combined 44,160-square-metre footprint is recalculated from concept geometry. Without an existing-building survey, property data, or regulatory controls, it cannot become a development quantum or demolition list. [data:geometry/buildings.geojson#BLDG-A-01] [metric:buildingfootprintareasqm]

Retain-renovate-demolish decisions follow “survey first, classify second, decide last.” Safe structures with historical or community value are retained first. Buildings that can be made accessible, efficient, and useful through insertion are renovated first. Only demonstrated structural or public-safety failures proceed to professional demolition review. New construction must show that existing space cannot meet the need and that a reversible pilot has passed evaluation. Every concept building is marked ‘pendingexistingbuildingsurvey’; no conclusion is made about a real building. [depth:retainrenovatedemolish] [depth:heightmassingcharacter]

The anti-gentrification toolkit has four institutional components: a resident-retention registration protocol, priority return for small businesses, temporary trading space during construction, and reinvestment of public benefit into community services. Real retention and return rates require a lawful, informed-consent baseline and longitudinal records. This package reports only the spatial support share and does not market a spatial proxy as a social outcome. [metric:residentretentionrate] [depth:developmentintensitycontrols]

Transport, Rail, Municipal Infrastructure, and Public Services

The transport system recognizes people and machines as distinct right-holders, while life safety, accessibility, and the right to refuse automation take precedence. The human network includes the reskilling corridor, night safety connection, continuous age-friendly walking, and screen-free green space. The machine network includes the controlled silicon-right-of-way corridor, a conceptual low-altitude layer, city-API nodes, and embodied-AI test grounds. Intersections use visible status, low-speed zones, human takeover, and incident shutdown; “optimal efficiency” cannot overrule pedestrians. [data:geometry/roads.geojson#ROAD-A-NIGHT] [depth:trafficrailslowparking]

The low-altitude concept-corridor length is recalculated from geometry, but airspace, clearance, noise, privacy, fire, route, and permitting data are absent. It is therefore a vertical functional layer for further study, not an engineering alignment. Silicon right-of-way likewise describes a controlled testing protocol, not a grant of road rights. No line on a figure is presented as an approved route or operating service. [metric:lowaltitudeconceptcorridorlengthm] [metric:siliconrightofwaylengthm]

Compute-energy coordination is a front-end spatial gate, not a late-stage mechanical optimization. New or expanded intelligent-computing facilities must first comply with current Beijing PUE requirements. The proposal recommends 30% green electricity as an initial design baseline verified by power-purchase credentials and metering, while expressly distinguishing it from a statutory project value. Recovered heat may connect to community energy only when temperature grade, seasonal demand, heat network, metering, and economics are demonstrated. [source:BJ-COMPUTE-2024] [metric:operationalpue] [metric:greenelectricityshare]

Public services use dual channels. AI may assist translation, queuing, accessibility navigation, and work-order recommendations, while staffed counters, telephone service, and on-site assistance remain available. A municipal agent can recommend but cannot autonomously issue a planning permit, interrupt water or power, exercise enforcement, or decide personal rights. Interface logs, permission revocation, continuity, and disaster-recovery drills are acceptance requirements. [source:BJ-PUBLIC-DATA-2026] [depth:municipalnewinfrastructure]

FIG. 04

HUMAN-MACHINE MOBILITY AND BLUE-GREEN RESILIENCE

Concept corridors; J3 human takeover precedes automated testing, not approved rights-of-way.



Derived from geometry/*.geojson and structured evidence; the ordinary-person journey, spatial interfaces, acceptance atlas, and reviewer navigation perform only offline reference or path checks and expected-FAIL negative tests; missing evidence remains unknown or conceptual.

Blue-Green Network, Public Space, and Urban Character

The blue-green network is not a backdrop for an AI demonstration. It is infrastructure for heat waves, intense rainfall, mental recovery, and ordinary social life. A conceptual sponge-resilience band along the Xiaoyue River wing connects modelling, inspection, and public risk information. Three Screen-Free Stillness Gardens prohibit mandatory interactive screens, targeted advertising, and emotion recognition, preserving space that is not continuously computed for knowledge workers, older people, children, and residents. [data:geometry/greenspace.geojson#GREEN-C-SPONGE] [metric:noscreengreenareasqm]

Design green space totals approximately 1,269,256.687 square metres, or 11.12% of the provisional site. Public-space polygons total approximately 121,447.158 square metres, or 1.06%. These values belong to design layers and are not a statutory green ratio, park implementation area, or public-space dedication. Official boundaries, river blue lines, heritage controls, and an existing-green survey would trigger regeneration and recalculation. [metric:greenratio] [metric:publicspaceration] [depth:bluegreenpublicspace]

Urban character uses “rail, time, interface” as its language. Rail represents the continuity of the century-old Jingzhang corridor, time marks the sequence of city releases, and interface nodes reference Zhongguancun’s open-source culture. Buildings and public facilities use removable, repairable, legible components. Screens, luminous façades, and corporate logos are not treated as an AI architectural style. Until formal protection information is available, historic resources, the former Tsinghuayuan Station, and park limits receive conservative avoidance. [standard:MOHURD-URBAN-DESIGN-MEASURES] [depth:heightmassingcharacter]

Three pilgrimage landmarks make the operating philosophy visible. The Human Release Hall publishes city issues, changes, and withdrawals. The Rail Interface Clock places railway time beside algorithm versioning and records who changed the city and when. The Screen-Free Recovery Beacon uses shade, wind, sound, and night safety to create a non-screen civic landmark. All three are original concepts and use no corporate trademark, personal likeness, or restricted imagery. [source:AGENT-TASKBOOK] [depth:riskmissingdata]

Renewal Projects, Implementation Policy, and Phasing

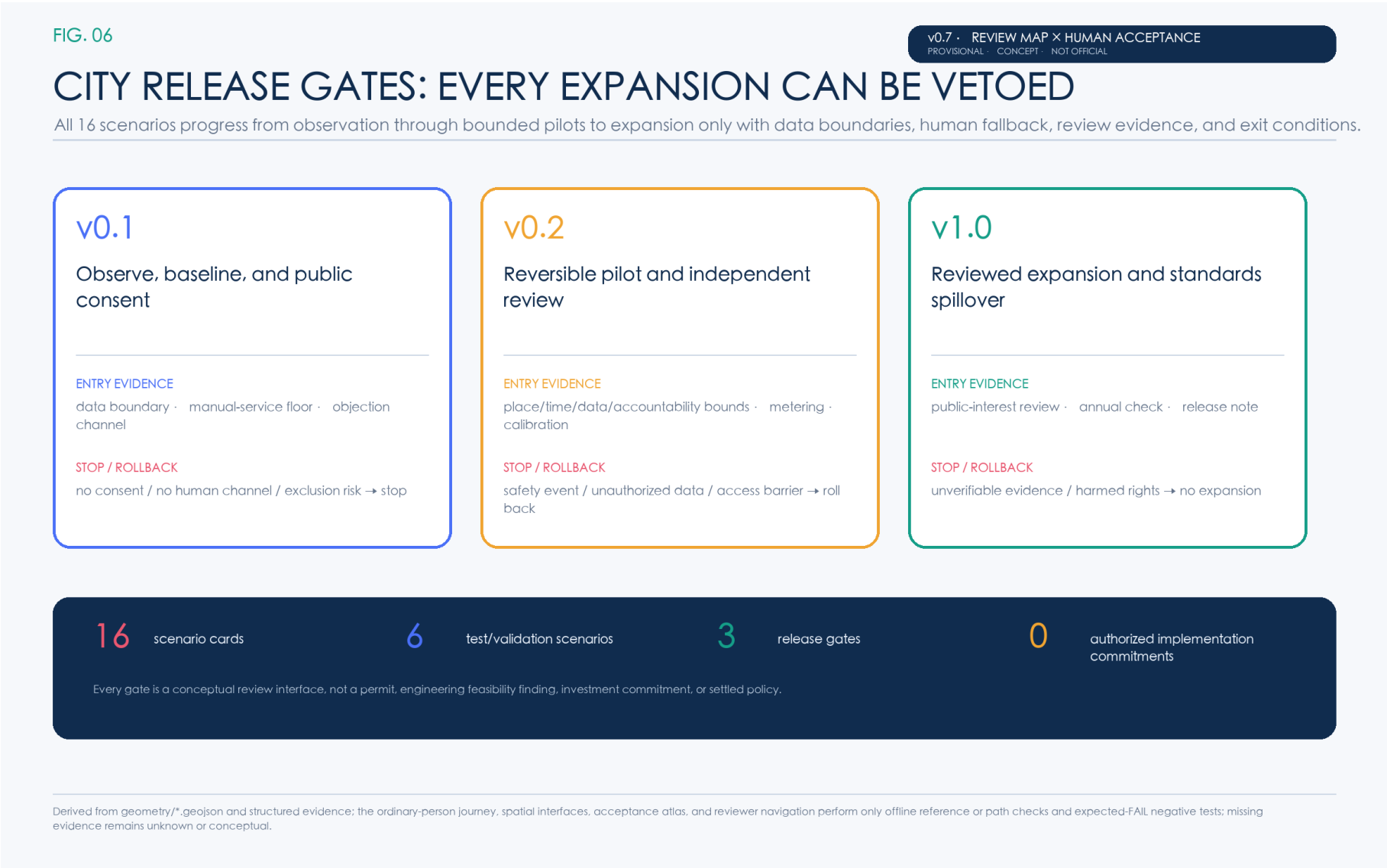
Implementation uses three city releases instead of a single completion date. Release v0.1 establishes baselines, a public issue list, and the minimum human-service guarantee. Release v0.2 conducts reversible pilots, independent evaluation, and incident drills. Release v1.0 expands only projects that pass public-interest, energy, climate, accessibility, and reversibility reviews. The three geometries cover the provisional site as governance responsibility zones, not as promised development timing. [data:geometry/phasing.geojson#PHASE-V02] [metric:versionreleasecount] [depth:phasingimplementation]

Before a suggested project enters its next gate, it leaves a synthetic human-city handoff receipt. H0 hands a problem from community observation to data and spatial review; H1 confirms an equivalent human entrance; H2 checks minimum fields and revocation; H3 reviews energy, climate, mobility, and safety evidence; H4 performs independent replay and opens an objection window; H5 records version differences and exit. The three cases are synthetic fixtures and the roles are abstract suggestions. They make who takes over, what permits acceptance, when to stop, and how to return to the previous gate queryable without presenting a suggested role as a real operator. [data:visual/assets/human-city-receipt.json] [data:visual/assets/implementation-operation-matrix.json]

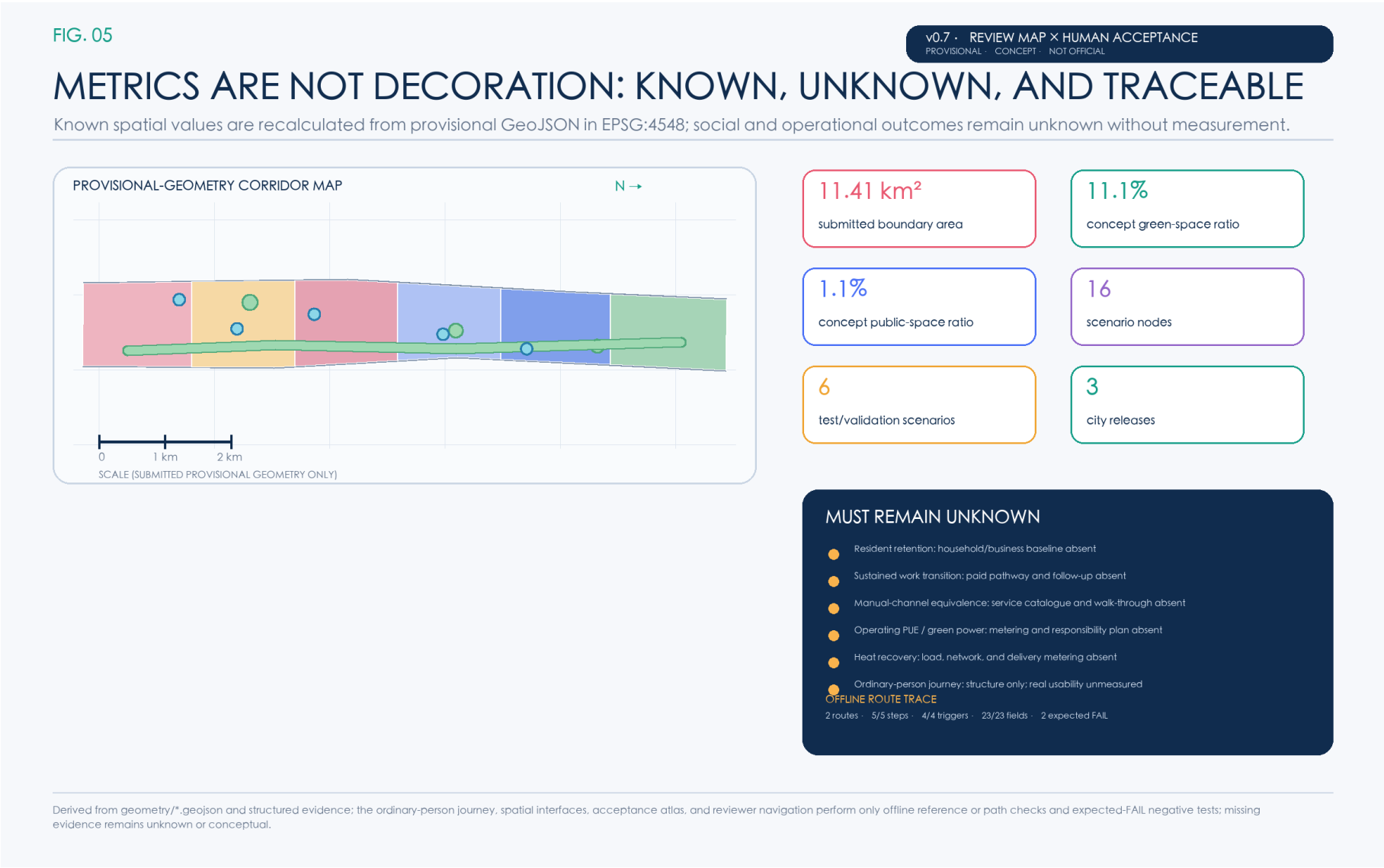
Project	Release	Prerequisite	Stop or rollback condition
Community-retention baseline and return protocol	v0.1	Resident consent, eligibility, and privacy rules	Do not publish a rate if a credible baseline cannot be formed
Reskilling Relay	v0.1	Employment partners, paid mechanism, and job tracking	Correct the program if it counts attendance without a job path
Human service and intergenerational learning	v0.1	Staffed service and accessibility walk-through	Stop digital migration when a human channel is unavailable
Screen-Free Stillness Gardens	v0.1	Confirm green/heritage boundaries	Remove any mandatory collection or targeted advertising
City API Exchange	v0.2	Data catalogue, permissions, logs, and exit	Shut down on unauthorized access, opacity, or missing takeover
Silicon-right-of-way test	v0.2	Traffic review, insurance, and incident plan	Withdraw if safety or public acceptance thresholds fail
Flood and sponge simulation	v0.2	Hydrology, blue-line, drainage, and model calibration	No expansion without calibration against events or drills
Compute-energy/recovered-heat audit	v0.2	IT load, PUE, green-power, and heat-demand evidence	No community-heat claim without metering or a heat user
Public Data Authorization Room	v0.2	Lawful authorization, public-interest priority, third-party evaluation	Close when privacy, trade-secret, or exit requirements fail
OPC Shared Laboratory	v0.2	Fair application, conflict disclosure, resource audit	Redesign if it becomes a closed club or tied-vendor channel
Standards Spillover Studio	v1.0	Open protocol and cross-district review	Never call an unvalidated output a national or industry standard

The core governance artifact is the release note. It records additions, changes, withdrawals, incidents, public objections, data permissions, and the next release's gates. The annual health check reviews space together with resident retention, job transition, energy, climate, data, and human channels. The public can file issues, and the maintenance team must answer “accept,” “test,” “reject,” or “await evidence,” with reasons. [depth:renewalprojectlist] [depth:phasingimplementation]

To avoid treating display as an upgrade, v0.2 added entry, advance, and exit gates to the three conceptual phase polygons, with the then-complete ledger of sixteen scenario cards as offline structured attachments; v0.8 adds SC-D04 and v0.9 adds the handoff receipt, bringing the current registry to seventeen cards. They require human takeover, authorization, insurance, metering, calibration, accessibility, and public review to be prerequisites instead of allowing any one technical effect to justify expansion. [data:geometry/phasing.geojson#PHASE-V02] [standard:MOHURD-URBAN-DESIGN-MEASURES]



`compliancematrix.json` covers twenty-three requirements across announcement groups 1.3, 1.4, 1.5 and agent.1 through agent.6. `standardmatrix.json` covers the announcement, taskbook, and three formally usable professional standards. `designdepthmatrix.json` covers fifteen required depth topics, while `selfcheck.json` stores the four gates. v0.2-v0.5 additionally provides ledgers for scenarios, release gates, persona fairness, implementation roles, metric recalculation, rights, bilingual equivalence, accessibility, data readiness, and spatial interfaces so professional teams can review evidence without treating appendices as proof of implementation. [standard:MOHURD-URBAN-DESIGN-MEASURES] [depth:metricsrecalculation]



International cases are JTC Punggol Digital District, Forum Virium Helsinki Smart Kalasatama, Barcelona City Council 22@, Seoul AI Hub, Korea Smart City Busan Eco Delta City, and Waterfront Toronto Quayside. They support comparative mechanisms only; they do not demonstrate that Jingzhang has identical legal, physical, economic, or institutional conditions. [source:CASE-BARCELONA22] [source:CASE-SEOUL-AI-HUB] [source:CASE-BUSAN-EDC]

The structured evidence entry points are `metrics.json` for recalculated metrics, `assumptions.json` for unknowns and confidence, `sources.json` for source boundaries, `geometry/` for spatial evidence, the requirement/standard/depth matrices, and `selfcheck.json` for the four gates. When official data changes, the City Release Room records the difference in a release note and reruns rendering, finalization, and self-check. [data:geometry/phasing.geojson#PHASE-V10] [depth:metricsrecalculation]